Human Genetics
https://doi.org/10.1007/s00439-019-02034-4

ORIGINAL INVESTIGATION

Missing heritability of complex diseases: case solved?

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Received: 2 November 2018 / Accepted: 28 May 2019
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Abstract
About 10 years ago, after the first large-scale genome-wide association studies (GWAS) were conducted to find genes associated with common complex diseases, investigators were surprised to find that the amount of heritability explained by the significant hits was very low for almost all the studied traits. Indeed, when compared to heritability estimates expected from the observed trait concordance within families, the heritability explained by the associated variants was always much smaller, more than ten times smaller for some traits. There was thus a problem of "missing heritability" and different hypotheses were proposed to help find this "missing heritability". These hypotheses involved novel research strategies in which different groups engaged including among others increasing sample sizes of GWAS or looking for rare variants and structural variations that were not captured by the SNP-chips used in GWAS. How successful have these efforts been in finding the "missing heritability"? Could it be that the problem of "missing heritability" was ill-defined? These are the questions that will be addressed in this paper by taking some different examples of complex traits.

Introduction

Finding the genes involved in the susceptibility to common diseases is one of the major challenges of genetic epidemiology. Over the years, technological progress has allowed a better characterization of the human genome variability and opened-up new possibilities and hopes to understand the genetic component of common diseases. First, back in the 1980s with the publication of the first microsatellite map of markers on the human genome, many linkage analyses were performed that searched for marker alleles that are transmitted together with the disease in multiplex families. Although the strategy was efficient to find the genes involved in monogenic diseases, it was disappointing in common diseases with very few significant hits found (Guo and Lange 2000). However, different measures of the contribution of genetic variants to disease risk such as sibling recurrence risks suggested that genetic factors were involved and should therefore be detected. Questions then arose on the nature of these genetic factors and it was suggested that, perhaps, they could not be detected by linkage analysis because they were too frequent in the population and with only modest effects (Altmuller et al. 2001; Thomson 2001). Alternative strategies were proposed that rely on association testing at a large scale (Risch and Merikangas 1996). This was not possible at that time at the scale of the entire genome but rather on candidate genes. Indeed, only few markers were available over the genome in the late 1990s and it became evident that efforts should then be made on the characterization of the common variants over the genome. The HapMap project was launched in 2002 to chart common genetic variation within the human genome and describe patterns of linkage disequilibrium (LD); i.e., the correlation that exist between these common variants in different populations worldwide (International HapMap 2002). Based on these data (International HapMap 2005), it was possible to define minimum sets of Single Nucleotide Polymorphisms (SNPs) that capture most of the common variation on the human genome and could be used to test for association at the scale of the entire genome. SNP-arrays were constructed using these so-called "tag-SNP" and genome-wide association studies (GWAS) became possible using these tag-SNPs as surrogates for all common variants in the human genome. The first GWAS was published in 2002 on myocardial infarction (Ozaki et al. 2002). It was followed by other early successes on macular degeneration (Dewan et al. 2006) that were achieved on rather small samples and using rather sparse marker maps. These

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Published online: 04 June 2019

first results raised hope that the future of genetic studies was in performing GWAS and that the main genetic contributors to human common disease were common variants, the so-called "common disease–common variant paradigm" (Reich and Lander 2001). A plethora of GWAS were performed since 2006 that found many reproducible hits for different diseases and traits. In April 2018, the GWAS catalog that provides a publicly available curated resource of all published human GWAS findings celebrated its 10th anniversary and reported 68,000 SNP-trait associations from 5000 studies (Buniello et al. 2019). Besides all the catches, these fishing expeditions have fallen short of expectations. Most of the associations found concerned variants of unknown significance that increased disease risk at very low levels and are not very useful for clinical applications (Hindorff et al. 2009). Moreover, after the first harvest of GWAS, scientists were disappointed to realize that, for all the traits studied, the GWAS hits that achieved genome-wide significance levels were unable to explain the heritability predicted from traditional genetic epidemiology studies based on twin or familial recurrence data. There was a problem of "missing heritability" that needed to be solved and different strategies were proposed to solve it (Maher 2008; Manolio et al. 2009).

In this paper, after a brief recall of the notion of heritability and the different ways to measure it, we will discuss these different strategies and how fruitful they have been, taking as examples different complex traits. We will also question the statistical models underlying heritability measures and call for a reappraisal of these models in light of new biological knowledge regarding the control of gene expression.

Heritability: what does it measure and how is it measured?

Heritability measures the proportion of the phenotypic variance in a population attributable to genetic differences. The concept was introduced by Fisher and Wright almost a century ago (Fisher 1918; Wright 1920). It relies on the idea that the total phenotypic variance V_P in a population can be partitioned into two components, a genetic component V_G and an environmental component V_E. The broad sense heritability H^2 is then defined as the ratio V_G/V_P. In most studies, however, what is reported is a strict or narrow sense heritability h^2 that relies on some additional partitioning of the genetic variance into an additive component also known as the breeding value (V_A), a dominance component (V_D) that takes into account the interaction between alleles at the same locus and an epistatic component (V_I) that accounts for interactions between alleles at different loci. The narrow sense heritability h^2 is then the ratio V_A/V_P (see Visscher et al. 2008 for a review).

Heritability can be estimated using empirical data on phenotype correlations between relatives and different approaches have been proposed that compare these correlations between different types of relatives (Tenesa and Haley 2013). One traditional design focuses on twins and compares the phenotype correlation between monozygotic and dizygotic twins. Estimates can also be obtained by considering the different pairwise genetic relationships within a pedigree and using linear mixed models (Lynch and Walsh 1998). For dichotomous traits, these methods do not apply directly. An additional assumption is required of the existence of an unobserved quantitative liability phenotype that explains the observations on the studied discrete character through a threshold model where individuals are assumed to be affected when their liability exceeds this critical threshold level (Falconer 1965). Rather than the heritability of the binary trait, it is the heritability of this liability that is estimated (Genin and Clerget-Darpoux 2015b; Visscher and Wray 2015). The validity of heritability estimates from the liability model depends strongly on the liability distribution that depends on the genetic architecture. Very different heritability estimates can be obtained for a same disease by assuming different underlying genetic architectures. This was well illustrated for schizophrenia with very different heritability estimates obtained by assuming a single underlying gene or by assuming a polygenic model (for a discussion on the limits of the liability threshold model, see Benchek and Morris 2013; Genin and Clerget-Darpoux 2015b; Visscher and Wray 2015).

The different estimates of heritability presented above require information on genealogies and phenotypic measures on individuals of known relatedness. When genetic data are available, instead of using genealogical information, researchers can infer individual relatedness from their genetic data and correlate this inferred relatedness with phenotype similarity. Such an approach was proposed in the late 19990s [see for example Ritland 1996] but could not really be used before dense maps of markers spanning the entire genome were available. It was first used in breeding genetics to perform marker assisted selection (Haley and Visscher 1998). In humans, with all the GWAS performed on different traits, information is now also available on the association between genetic variants and phenotypes. It is therefore possible to directly quantify the contribution of these genetic variants to phenotypic variance and measure what is called "genomic heritability" (Yang et al. 2010; de Los Campos et al. 2015). Briefly, the idea is to infer the proportion of variance that can be explained by a linear regression on a set of markers used as explanatory variables. This was done first using only the significant SNPs associated with traits in GWAS and it was found that there existed a major gap between this genomic heritability and the heritability estimates from phenotype correlation studies between relatives.

A classic example is height which is an easy-to-measure trait in humans and used as a model to study the impact of genes on complex traits since the pioneer works (Fisher 1918). The heritability of height in humans was estimated to be around 80% based on its correlation between relatives. In 2008, three GWAS on height were published in the same issue of Nature Genetics that found, respectively, 27, 12 and 20 associated genomic regions (Gudbjartsson et al. 2008; Lettre et al. 2008; Weedon et al. 2008). Taken together the associated loci in each study only explained, respectively, 3.7, 2.0, and 2.9% of the adult population variation in height, almost nothing compared to the 80% predicted from early studies. Similar results were obtained for GWAS performed on different common diseases including for example autism, diabetes, heart diseases or Alzheimer's disease. For late-onset Alzheimer's disease for example, the heritability estimated based on the twin data from the Swedish Twin Registry was 58% with a very large 95% confidence interval ranging from 19 to 87% (Gatz et al. 2006). When taking into account all SNPs associated with the disease based on the different GWAS studies conducted, the genomic heritability was 7.78% (Ridge et al. 2013). It was even below the lower tail of the very large 95% confidence interval of the twin heritability. Investigators were then faced with the so-called problem of "missing heritability" (Maher 2008) and the chase to find this missing heritability began.

Chasing the missing heritability of complex diseases

To understand the problem of missing heritability from GWAS studies and explain where the missing heritability might be found, several hypotheses were proposed (Maher 2008; Manolio et al. 2009). We will review these different hypotheses and how they have been addressed since the problem of missing heritability was first described 10 years ago.

Many loci with small effects contribute to phenotype variance

A first explanation was that the GWAS analyses were not powered enough to capture all the genetic variants involved in disease susceptibility and that a lot of variants with small effects were missed. The significant hits from GWAS might then only be the visible parts of the iceberg. Genomic heritability estimates could, therefore, be improved by taking into account all the genetic data to incorporate smaller effects that did not reach significance but could, however, significantly contribute to phenotype variability. To test this hypothesis, Yang et al. (2010) reanalyzed GWAS data sets on height by fitting a mixed model that includes all the SNPs with SNP effects treated as random variables. They found that 45% of the variance in height can be explained by considering all SNPs simultaneously compared to less than 5% when only genome-wide significant SNPs were considered. The method was then extended to estimate genomic heritability of binary traits (case–control data) (Lee et al. 2011) and implemented in a user-friendly software (Yang et al. 2011). When applied to late-onset Alzheimer's disease data, the method yields a genomic heritability based on all SNPs of 33.12% that is fourfold the estimate obtained with only the significantly associated SNPs, showing that there might indeed be many additional SNPs with small effects that together play a significant role in the phenotype variance. This hypothesis is also corroborated by the fact that GWAS conducted on ever growing sample sizes is able to find new hits. For height, the number of associated variants has grown from ~40 in the first GWAS to ~700 when study sizes increased to 250,000 individuals and to 3290 in the latest study that included 693,529 participants (Yengo et al. 2018). For late-onset Alzheimer's disease (LOAD), novel loci were also detected when GWAS sample sizes were increased. For example, in the study by Lambert et al. (2013) by increasing the sample sizes to ~74,000 individuals, 11 new loci could be identified. However, the total number of LOAD associated loci remains much smaller than for height (Shen and Jia 2016) and several of them target the same biological pathway involved in the Aβ peptide processing (Nicolas et al. 2016).

More recently, it was proposed that rather than polygenic, the genetic architecture of common diseases could in fact be "omnigenic" with mainly all active genes affecting every complex traits (Boyle et al. 2017). Indeed, when looking at several different complex traits including height, schizophrenia, rheumatoid arthritis or Crohn's disease, Boyle et al. (2017) found that the signals identified by GWAS were located all over the genome and were not clustered around some disease specific genes. They recognized, however, the existence of some "core genes" specific to each disease but the hypothesis that such "core genes" play a significant role in complex disease is still debated (Wray et al. 2018).

Rare variants with larger effects play a significant role in common diseases

A second explanation for the missing heritability is that rare variants that are not captured by SNP-chips could be major contributors of common disease susceptibility. Indeed, population genetics models and the first resequencing studies have shown that rare variants that are only seen in very few individuals constitute the vast majority of population genetic diversity. An important proportion of these variants is mildly deleterious and thus good candidates for being involved in common disease susceptibility (Kryukov et al. 2007). An

alternative model for common disease genetic architecture was then proposed where it was assumed that there existed plenty of rare genetic variants each with high penetrance that could be involved in common disease susceptibility. It is the so-called "common disease-rare variant model" (Schork et al. 2009; Saint Pierre and Genin 2014). Under this model, an important genetic heterogeneity is expected and other study designs and statistical methods than the ones used in GWAS are needed to uncover the genes involved (Zuk et al. 2014; Auer and Lettre 2015).

There are several examples of rare and low-frequency variants associated with common traits. For human height for example, a study was recently performed using ExomeChip data, a genotyping array specially designed to search for low-frequency variants, on 711,478 individuals (Marouli et al. 2017). A total of 83 height-associated coding variants with minor allele frequencies in the range 0.1–4.8%, were identified. The effects of these rare variants on height were much greater than the effects of common variants, in the order of ten times greater than the mean effect size of common height alleles estimated in the same data set. Taken together, these 83 rare and low-frequency variants explained 1.7% of the variation in height in the UK Biobank data set. When adding these rare and low-frequency variants to the common variants associated with height, 27.4% of the variation can be explained. Interestingly, several of the rare variants identified (49 out of the 83) are located near common variants previously associated with height. However, a majority of these rare variants are independent of the previously described height SNPs. Some of these variants are located within the same genes and gene-based analysis was able to detect three novel genes associated with height that were not detected when rare and low-frequency variants were investigated one by one. Moreover, several of the coding variants revealed in this study were located in genes mutated in monogenic skeletal growth disorders, suggesting that there might exist some sort of continuum between monogenic and common traits—the authors used the term "allelic series" to refer to this gradient from familial penetrant mutations to common variants with mild effects.

In Alzheimer's disease, apart from the rare variants involved in the monogenic sub-entities of the disease, rare and low-frequency variants were also identified in the more common forms of the disease. The first gene identified with a low-frequency variant involved in LOAD was TREM2. There were then some other genes described including SORL1 or ABCA7 where a burden of rare variants was found among cases (for a review, see Nicolas et al. 2016). The contribution of these different genes to Alzheimer's disease susceptibility was evaluated in a French cohort of 852 EOAD, 927 LOAD and 1273 controls and it was found that rare protein-truncating and strictly damaging missense variants in SORL1, TREM2 and ABCA7 explained, respectively, 1.42%, 1.17% and 1.33% of the total EOAD heritability (Bellenguez et al. 2017). In comparison, using the same model to compute heritability, ApoE4 allele was found to explain 9.12% of EOAD heritability.

Another interesting example where contradictory results were obtained is type-2 diabetes (T2D). In a recent study on T2D and related quantitative traits reflecting glycemic control performed in 20 large Mexican–American pedigrees, Jun et al. (2018) did not detect any rare variants associated with these traits. They concluded that rare variants with large effect were unlikely to explain a large portion of T2D risk. Indeed, should such variants have explained ~33% of the variation in risk for T2D, they showed through simulations that their study had 80% power to detect at least one such variant. This is in good agreement with a previous study by Fuchsberger et al. (2016) where whole-genome and exome sequencing on large samples of cases and controls did not detect any rare variants associated with T2D. The authors concluded that rare variants were unlikely to play an important role in T2D susceptibility and that T2D-associated loci have not been under natural selection. This was, however, recently contradicted by a large meta-analysis that found 4 rare variants conferring risks for T2D that have considerable effects with odd-ratios between 1.33 and 2.74 (Xue et al. 2018). The authors estimated that negative selection is at play on deleterious rare alleles and that there are probably still many rare risk alleles to be discovered in T2D. Recent results using UK biobank data on different traits have also found that deleterious variants affecting complex traits are under the action of negative selection. This was the case especially for reproductive and cardiovascular traits that are closely related to fitness (Gazal et al. 2018; Zeng et al. 2018). Under this scenario, we can expect that many extremely rare variants are still to be found in complex diseases and alternatives strategies to classical association studies that search for signature of negative selection should probably be considered.

Structural variants are important contributors in common diseases that have been poorly explored

Like rare variants, structural variants also suffered from insufficient coverage on SNP-chips and they were completely ignored in the first association studies of complex traits. However, structural variants and Copy-Number Variants (CNVs) in particular, that are variable numbers of copies of some genomic regions of about 100 kilobases and greater, represent an important form of variation in the human genome, encompassing more nucleotides over the genome than SNPs. Methods and algorithm have been developed to call CNVs from SNP-chip data [see for example PennCNV (Wang et al. 2007) or QuantiSNPs (Colella et al. 2007)] and studies were performed to assess the contributions of CNVs

to complex traits (see McCarroll 2008 for a review). Among these different studies, the one from the Welcome Trust Case Control Consortium on 16,000 cases of eight common diseases and 3000 shared controls suggested that most of the common CNVs were well tagged by SNPs and unlikely to explain much of the missing heritability of complex traits (Wellcome Trust Case Control et al. 2010).

However, when analyzing SNP-chip data, no account is taken of allele dosage; i.e., the homozygous AA, hemizygous A and duplicative AAA genotypes are considered the same and collapsed into a single state (homozygous AA). It could explain why the contribution of CNVs to complex traits has been under-evaluated (Marenne et al. 2013; Gamazon et al. 2014). Indeed, when taking into account the actual allele dosage in genotypes, Gamazon et al. (2014) found an improvement in power to detect trait-associated loci and especially eQTL. Similar conclusions were reached by Marenne et al. (2013) based on simulations under different disease genetic architectures.

Recent CNV-association meta-analysis performed on anthropometric traits in 191,161 European adults from 26 cohorts found some rare but recurrent CNVs that contributed significantly to variations in these traits, including height (Mace et al. 2017). For this latter trait, increased CNV burden was associated with shorter stature with a higher impact of deletions than duplications. A 600 kp deletion on the 16p11.2 region was associated with decreasing height and increasing weight. This CNV association explained a substantial fraction of the association between the SNP near FLJ25404 and height. Apart from the 16p11.2 deletion, 3 other CNV regions were found to be associated with height on chromosomes 1, 3 and 11.

In Alzheimer's disease, CNVs have been found in all forms of the disease. In the mendelian EOAD, duplications in the APP gene have been found [see for example Rovelet-Lecrux et al. (2006)]. In autosomal dominant early-onset AD without mutations in known genes, several CNVs were found in different genes, a majority of which were involved in Abeta peptide metabolism (Rovelet-Lecrux et al. 2012). CNVs were also identified in sporadic AD by CNV-GWAS in different studies [for a review, see Cuccaro et al. (2017)].

Interactions between many factors are at play in most common diseases

Another hypothesis to explain missing heritability is the presence of gene–gene and gene–environment interactions. Indeed, a major assumption of heritability computation is the absence of interactions. Interaction refers here to statistical interaction; i.e., a departure from additivity of effects on a specific outcome scale (Dempfle et al. 2008). This is different from biological interactions where direct physical or chemical reactions exist between the factors. In complex diseases, complex biological interactions between genetic and environmental factors are at play. The effect of a genotype on phenotype often depends on the genetic background and on the genotypes at other loci (Carlborg and Haley 2004; Mackay and Moore 2014). One way to account for these effects in statistical models consists in adding interaction terms. Attempts to unravel gene–gene interactions have, however, often been unsuccessful because of a lack of power of most studies to detect statistical interactions. Indeed, very large sample sizes or effects are required if one wants to test for all possible interactions between genes while controlling for type-one error rates. Moreover, the variable coding scheme used to test for interaction is important and should be carefully chosen depending on the hypothesis to be tested (Cordell 2002; Aschard 2016). When considering gene–environment interactions, multiplicity is less an issue but a limitation comes from the fact that most environmental exposures are not constant over time. These exposures may then be difficult to measure, requiring the follow-up of patients and longitudinal data collection.

Different authors have explored the impact of departure from additivity on heritability estimates or risk prediction. In a simulation study, Aschard et al. (2012) found that the inclusion of gene–gene and gene–environment interactions do not significantly improve complex disease risk prediction. Similar conclusions were reached by Zhu et al. who developed a method to estimate the respective proportions of phenotype variance that could be attributed to additive and dominance genetic variation at common SNPs. They found that the latter contributes very little to phenotype variation and does not explain the missing heritability (Zhu et al. 2015). This was in contradiction with previous work by Zuk et al. (2012) that suggested that ignoring statistical interactions can lead to an inflation of the expected heritability. This inflation could then explain some of the differences between the SNP-based heritability and the heritability measured by the phenotypic correlations between relatives. Indeed, when measuring the proportion of heritability explained by known genetic variants, one computes a ratio $\pi_{\text{explained}}$ between the proportion of phenotypic variance explained by the additive effects of known variants, h^2_{known}, in the numerator and the proportion of phenotypic variance attributable to the additive effects of all variants including those not yet discovered, h^2_{all}, in the denominator. This latter quantity h^2_{all} is not known but inferred from the observed phenotypic correlations in a population, h^2_{pop}, and h^2_{pop} is only a good estimate of h^2_{all} in the absence of gene–gene and gene–environment interactions. In the presence of interactions, h^2_{pop} can significantly exceed h^2_{all}, creating what they call "phantom heritability" and $\pi_{\text{explained}}$ is then underestimated. As an example, in Crohn's disease, if interactions are ignored and we assume $h^2_{\text{pop}} = h^2_{\text{all}} = 50\%$, the 71 major risk

loci explained $h^2_{known} = 10.8\%$ of the phenotypic variance and $\pi_{explained} = 21.5\%$. Now, assuming a model where interactions exist between say 3 genes then, the true total heritability turns out to be $h^2_{all} = 18.6\%$ instead of 50% and $\pi_{explained}$ is then much greater, reaching 57.5%.

In a recent study, Kanoungi and Nothnagel (2018) considered other examples of genetic architectures that depart from additivity: gene-regulatory networks where genes and transcription factors interact with each other to control gene expression level. Using extensive time-forward simulations, they studied the impact of different pathway structures on allelic spectra and epidemiological characteristics and compared these results to the ones expected under the classical liability threshold additive model. They found important departures especially under models where serial connections existed between the components and functionality was only achieved when each of the components were functional.

Finally, several authors have also pointed out that heritable epigenetic factors could also account for some important parts of the missing heritability (see for example Koch 2014; Trerotola et al. 2015). Bourrat et al. compared the way epigenetic factors influence heritability estimates in traditional quantitative genetics and when using GWAS data. They found that the former estimates account for the effects of epigenetic factors while the latter do not. They claimed that these differences could explain a substantial part of the missing heritability (Bourrat et al. 2017). Other investigators have also suggested that the microbiome composition and functional diversity are other important factors that could also explain the missing heritability as they are not considered in the heritability estimates derived from SNP data (Sandoval-Motta et al. 2017).

Revisiting the missing heritability problem

From the methodological viewpoint

Apart from the problem of gene–gene and gene–environment interactions described above, some limitations of the methods used to compute genomic heritability have been identified and solutions proposed to solve them (for a review, see Lee and Chow 2014; de Los Campos et al. 2015). First, population stratification can bias heritability estimates and, depending on the phenotype, adjustment on top principal components (introduced as fixed effects in the model) does not fully solve the problem (Browning and Browning 2011; Dandine-Roulland et al. 2016). Estimates could also be biased if local differences in linkage equilibrium (LD) are not accounted for. Indeed, causal SNPs located in regions of high LD are then given too much weight and it was therefore suggested to weight the SNP contribution according to local LD (Speed et al. 2012). Alternatively, LD can be used to distinguish between the true polygenic effects and biases due to population stratification using LD score regression (Bulik-Sullivan et al. 2015). Minor allele frequency also seems to be an important parameter to take into account for heritability estimation (Speed et al. 2017; Gazal et al. 2018).

Methods based on restricted maximum likelihood (REML) such as GCTA that estimate the heritability explained by all SNPs including those with small effects were found to considerably underestimate heritability when applied to case–control data (i.e., binary traits) because this sampling scheme induces GxE interactions (Golan et al. 2014). This downward bias depends on many parameters and was found to increase with increasing sample sizes. To correct for this bias, Golan et al. proposed a novel method referred to as the phenotype correlation genotype correlation (PCGC) regression that is based on the Haseman–Elston regression method where the heritability is simply the slope of the regression of phenotypic correlation between individuals onto their genotypic correlation. Using the PCGC method on six different diseases, the authors found that common variants explained between 41 and 68% of the heritability of these traits.

More recently, other methodological developments have also been proposed. For example, Young et al. introduced a novel method to estimate heritability based on identity-by-descent that is less sensitive to environmental bias. Applying the method to data on 14 different traits available on 54,888 Icelanders, they found that environmental effects, when not properly accounted for, could lead to an inflation of heritability estimates. This was true in particular for educational attainment which heritability could be inflated by a factor of 1.6 (Young et al. 2018). A comparison of several of the methods available to estimate genomic heritability was published by Evans et al. (2018). Extensive simulations were performed to compare these different methods and provide guideline to help interpret heritability estimates.

From the conceptual viewpoint

Some authors, especially in the field of behavioral genetics, have strongly criticized the abuse of heritability measures in the genetic literature. They believed these measures have no real utility when considering complex traits and they are often badly interpreted [see for example Moore and Shenk (2017)]. Indeed, it is important to recall that heritability is not a measure of how genetic a trait is. It is a measure of "how variation at the DNA level relates to variation in traits across a population". It strongly depends on the population where the traits are measured and it is thus not at all a universal measure that could be transposed to any population context. Moreover, it has no relevance at the individual level.

Another false assumption about heritability is the idea that the genetic influence on trait development can be

separated from the environmental context. Indeed, the theory of heritability was developed at a period of time when geneticists had a very deterministic view of the influence of genes on traits and believed that genes came first to determine phenotypes and that the environment only had a small "finishing" impact (Lush 1937). However, this is a misleading conception as we now know that gene expression depends on many factors including environmental factors and complex interactions between these factors. It is therefore not possible to distinguish and quantify the influence of one factor independently of the others. This is an old debate that recently came back as "nature" versus "nurture". In the analysis of complex traits, however, "just as we cannot say how much of the area of a rectangle is due, separately, to each of its two dimensions", it is not possible to separate "nature" from "nurture" (Lewontin 1974; Feldman and Ramachandran 2018). Even if we add interaction terms in a statistical model, we cannot really capture the complex relation that exists between these factors.

Moreover, focusing on the genetic factors only, assuming an additive polygenic model underlying the liability of multifactorial diseases is also highly questionable. It is overly simplistic as is the idea that the genetic effects could be captured by the individual analysis of single nucleotide polymorphisms. We warned against the "dramatic resurgence of the GIGO (Garbage-In Garbage-Out) syndrome in genetics" that came from misconception and abuse of the concept of heritability and the adoption of the polygenic additive model as a universal model for complex diseases (Genin and Clerget-Darpoux 2015a, b).

Conclusion

Heritability estimates have flourished in the human genetics literature over the last 10 years following the development of methods based on GWAS data that allowed quantifying the heritability explained by SNPs. Comparing these estimates against the ones obtained using traditional epidemiological measures based on phenotype concordance between relatives has been the new game of this last decade since it was first shown that SNP-based estimates were significantly lower and that there was some "missing heritability". In each new study on complex traits that report novel genetic associations, heritability needs to be computed to show how much closer to the reference heritability we are. We have illustrated this phenomenon for both a quantitative trait, human height, and a complex disease, Alzheimer's disease. Height is the "reference" trait for heritability estimates as it is an easily measured trait that served as model in quantitative genetics. Alzheimer's disease is a complex disease that has strongly benefited from genetic discoveries as genetics has helped understand the physiopathology of the disease. We showed how technical developments that improved our knowledge of the human genome have allowed the identification of new genetic players and how it has improved heritability estimates. However, we also questioned these heritability estimates. What is their usefulness and what do they really mean?

Statistical methods to estimate heritability from genomic data have improved and new methodological developments are constantly proposed to capture maximum information from the data. People are debating on whether and how heritability estimates can further be refined. Inferences on the genetic architecture of complex diseases are made based on these estimates derived from a model, the polygenic additive model, that was proposed by Fisher a century ago when genetics was at its infancy (Nelson et al. 2013). This model works rather well for predicting the breeding values of animals in experimental settings where the environment is kept constant and controlled. In natural populations, however, it leads to heritability estimates that are difficult to interpret and often have very large confidence intervals (note, however, that these confidence intervals are barely reported).

Complex diseases are driven by many genetics and environmental factors that interact through complex networks to lead to the observed phenotype. Heritability measures aim at quantifying what in the phenotype variability is due to the genes and what is due to the environment. But, can we really make this distinction and can we learn about the genetic architecture of complex diseases through heritability quantification based on simplistic models that ignore interactions and try to dissect inseparable components? Many geneticists have warned about the abuse of heritability and call for the need to go beyond simplistic models and consider diseases in their complexity.

We believe that the "missing heritability problem" is an ill-posed problem. Solving it by refining over and over again statistical models derived under the "polygenic paradigm" or using more and more sophisticated sequencing machines will not answer the fundamental and biological question of why that individual is affected by the disease and that other individual is not. It is that case that needs to be solved and not the "missing heritability" case.

Compliance with ethical standards

Conflict of interest The author declares that they have no conflict of interest.

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